Ch 3.1-2: (Multi)-Linear Regression

Lecture 5 - CMSE 381

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Fri, Sep 8, 2023

Announcements

Last time:

- Started 3.1 - Single linear regression

Announcements:

- Office Hours: Tues - Fri
- Homework #1 grades and feedback posted
- Homework #2 Due Mon, Sep 11

Covered in this lecture

- Confidence interval, hypothesis test, and p-value for coefficient estimates
- Residual standard error (RSE)
- R squared
- Setup for multiple linear regression

Section 1

Last time

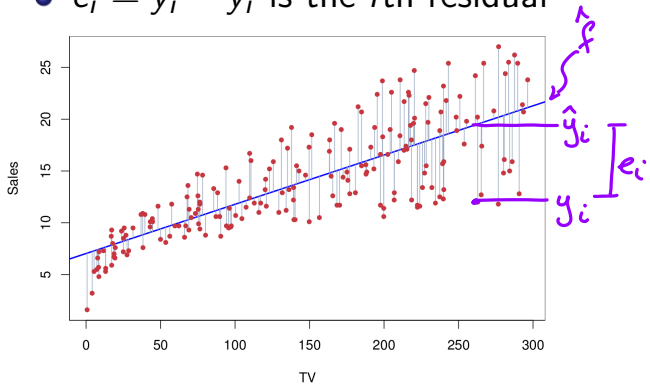
Setup

- Predict Y on a single predictor variable X

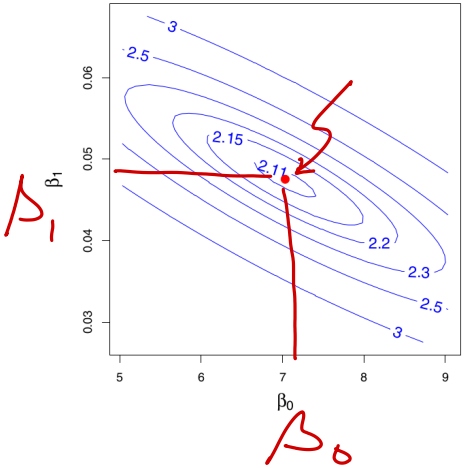
$$Y \approx \beta_0 + \beta_1 X$$

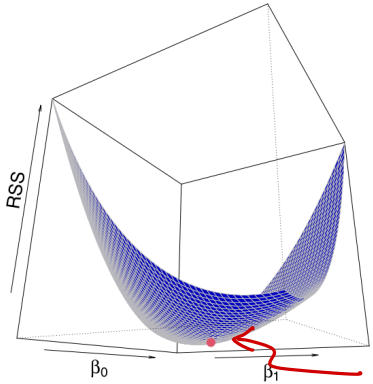
- "$\approx$" .... "is approximately modeled as"

- Given $(x_1, y_1), \cdots, (x_n, y_n)$
- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be prediction for Y on ith value of X.
- $e_i = y_i - \hat{y}_i$ is the ith residual



Least squares criterion: RSS





Residual sum of squares RSS is

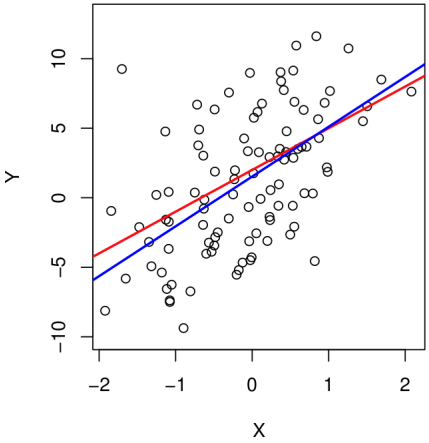
$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

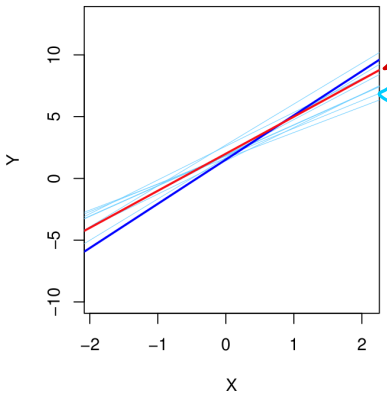
[redacted]

Find β_0 and β_1 that minimize the RSS.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$
$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

Linear regression is unbiased





real f

all the $\hat{f}$

100 data pts from

$y = 2 + 3x + \varepsilon$

Normal $\mu = 0$

Calculate $(\hat{\beta}_0, \hat{\beta}_1)$

Section 2

New stuff on evaluating models

R^2 vs p-value

Variance of linear regression estimates

- Variance of linear regression estimates:

$Y = f(X) + \varepsilon$

Standard error

$$\mathrm{SE}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$ → Irreducable error

- Residual standard error is an estimate of σ

$$RSE = \sqrt{RSS/(n-2)}$$

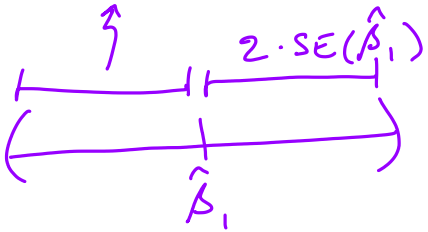
# of data points

$$RSS = \sum_{i=1}^{n} e_i^2$$

Confidence Interval

The 95% confidence interval for β_1 approximately takes the form

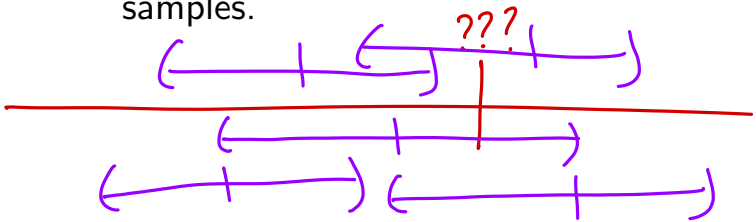
$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$$



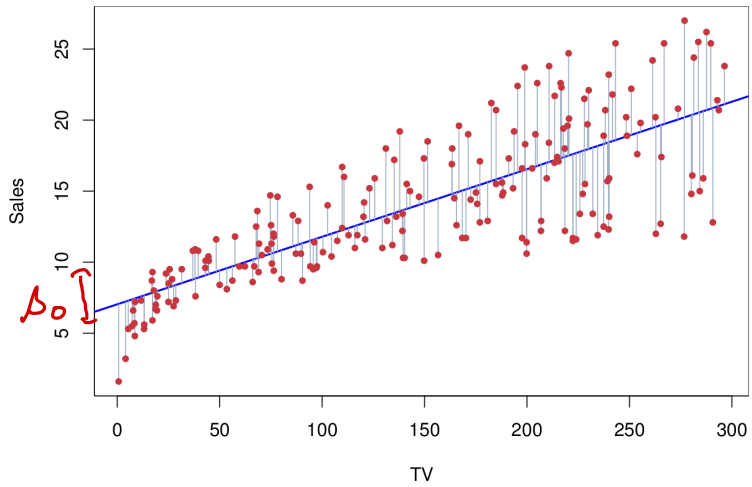
Interpretation:
There is approximately a 95% chance that the interval

$$\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$$

will contain β_1 where we repeatedly approximate $\hat{\beta}_1$ using repeated samples.



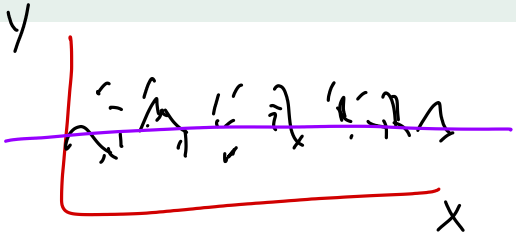
CI in Advertising data



For the advertising data set, the 95% CIs are: (approximately)

- β_1 :: $[0.042, 0.053]$
- β_0 :: $[6.130, 7.935]$

Hypothesis testing



null hypothesis
read "H-naught"

H_0: There is no relationship between X and Y $\rightarrow$ $H_0: \beta_1 = 0$

$f(x) = \beta_0$

H_A: There is some relationship between X and Y $\rightarrow$ $H_A: \beta_1 \neq 0$

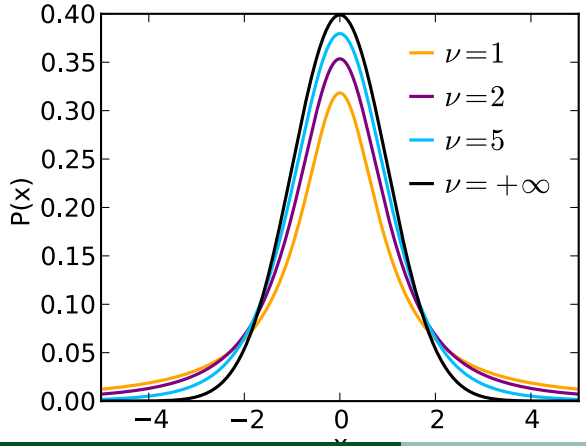
Alternative hypothesis

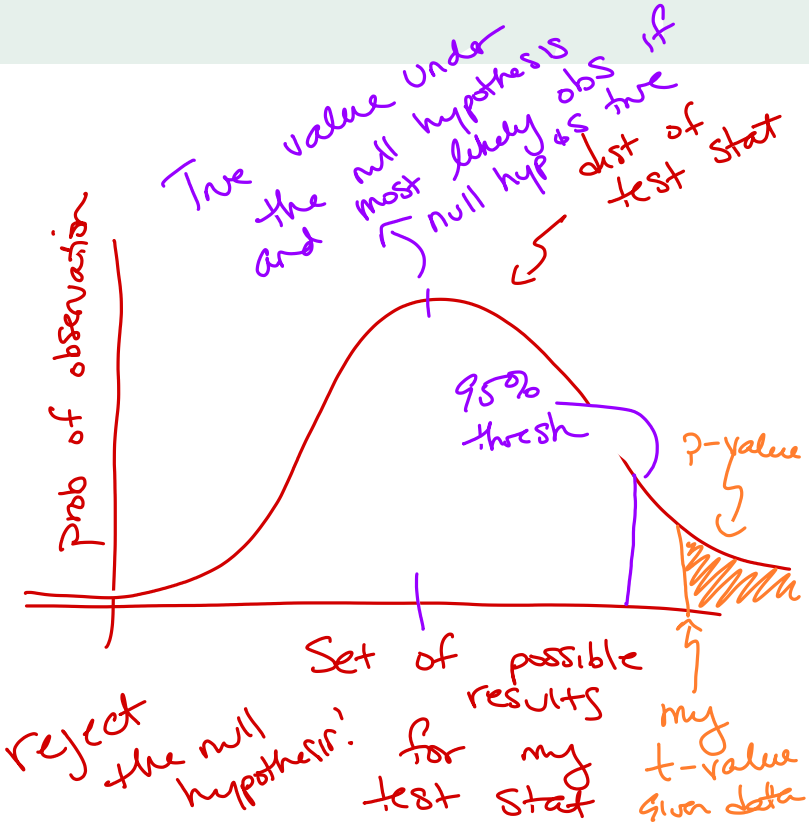
Test statistic and p-value

Test statistic:

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$$

t-distribution with $n - 2$ degrees of freedom





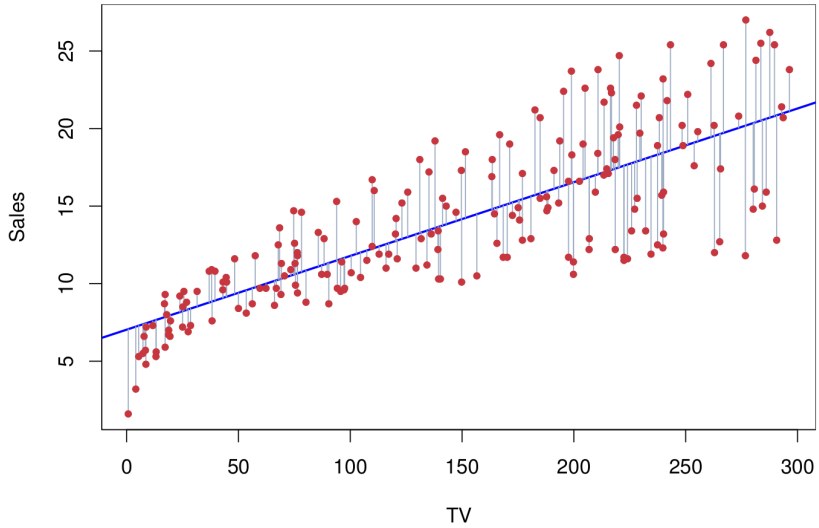
Advertising example

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

very small p-value (< 0.05)

reject the null hypothesis

$Y \approx \beta_0 + \beta_1 X_{TV}$



Assessing the accuracy of the module: RSE

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-2} RSS}$$
$$= \sqrt{\frac{1}{n-2} \sum_i (y_i - \hat{y}_i)^2}$$

Quantify the extent to which the model fits the data

- Estimate the std dev of ε
- Avg amount the response will deviate from true regression line

Assessing the accuracy of the module: R^2

Can only be in $[0, 1]$

RSS is the amt of variability after regression

$\sum_{i=1}^{n} e_i^2$

R squared:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

where total sum of squares is

$$TSS = \sum_i (y_i - \bar{y})^2$$

total variance in the respose of Y, variability before doing regression

R^2 is the proportion of variability in Y that can be explained using X

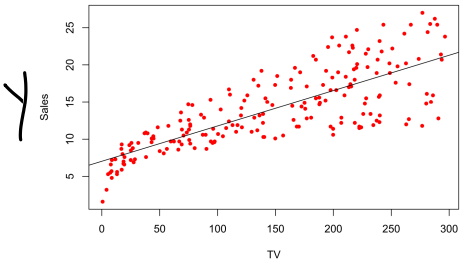
- If $R^2 \approx 1$, then $\frac{RSS}{TSS}$ is small ~ good job = small RSS

large proportion of variability is explained by regression

If $R^2 \approx 0$, then $\frac{RSS}{TSS} \approx 1 \Rightarrow RSS \approx TSS$

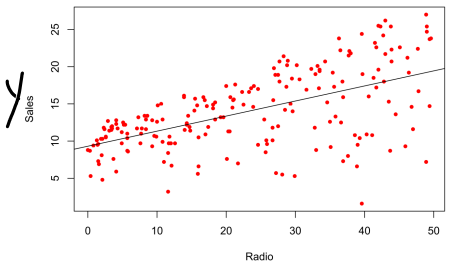
regression does not explain much of variability

Advertising example



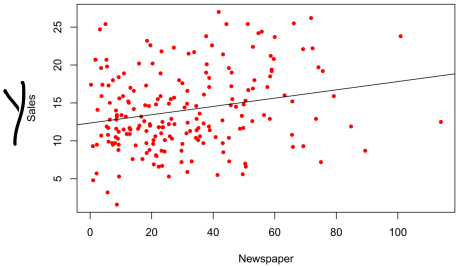
TV

$R^2 = 0.61$



Radio

$R^2 = 0.33$



News

$R^2 = 0.05$

Coding group work

~~Run the section titled "Assessing Coefficient Estimate Accuracy"~~

Do 1st half

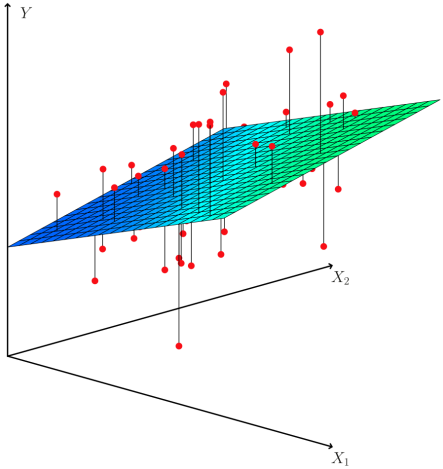
Section 3

Multiple Linear Regression

Setup

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p + \varepsilon$$

β_j is the avg effect on Y of one unit increase on X_j



Estimation and Prediction

Given estimates $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \cdots, \hat{\beta}_p$, prediction is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

Minimize the sum of squares

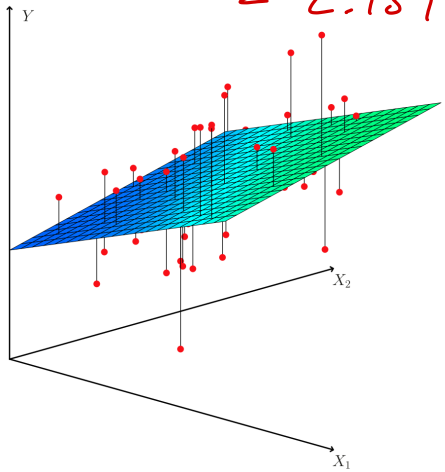
$$\begin{aligned} RSS &= \sum_i (y_i - \hat{y}_i)^2 \\ &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_1 - \cdots - \hat{\beta}_p x_p) \end{aligned}$$

Coefficients are closed form but UGLY

Advertising data set example

$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \texttt{TV} + \beta_2 \cdot \texttt{radio} + \beta_3 \cdot \texttt{newspaper}$$

$= 2.939 + 0.046\,TV + 0.189\,radio - 0.001\,News$



	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

Interpretation of coefficients

$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \texttt{TV} + \beta_2 \cdot \texttt{radio} + \beta_3 \cdot \texttt{newspaper}$$

(units of \$1,000)

	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

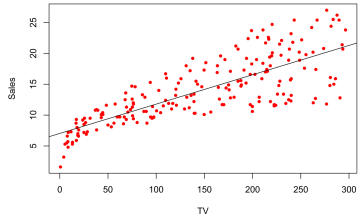
Sales in units of 1,000

Fix TV and newspaper spending

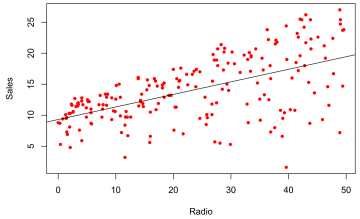
↳ Spending \$1K more on radio results in 189 additional units sold

Newspaper basically no relationship

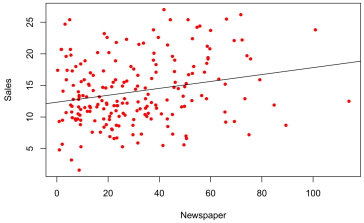
Single regression vs multi-regression



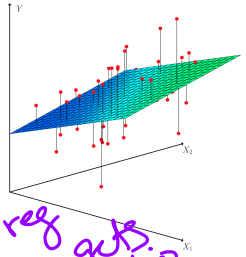
	Coefficient
Intercept	7.0325
TV	0.0475



	Coefficient
Intercept	9.312
radio	0.203



	Coefficient
Intercept	12.351
newspaper	0.055



	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

In simple reg newspaper gets credit for radio

Correlation matrix

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

high

Markets w/ lots of radio also have lots of newspaper

Coding group work

Run the section titled "Multiple Linear Regression"

Next time!

Next time

Lec #	Date			Reading	Homeworks	***Quizzes*** *(Note: These are not announced until after they happen)*
1	Mon	Aug 28	Intro / First day stuff / Python Review Pt 1	1		
2	Wed	Aug 30	What is statistical learning?	2.1		
3	Fri	Sep 1	Assessing Model Accuracy	2.2.1, 2.2.2	HW #1 Due	Quiz #1
	Mon	Sep 4	No class - Labor day			
4	Wed	Sep 6	Linear Regression	3.1		
5	Fri	Sep 8	More Linear Regression	3.1/3.2		
6	Mon	Sep 11	Even more linear regression	3.2.2	Hw #2 Due	
7	Wed	Sep 13	Probably more linear regression	3.3		
8	Fri	Sep 15	Linear regression coding module			
9	Mon	Sep 18	Intro to classification, Logisitic Regression	2.2.3, 4.1, 4.2, 4.3		
10	Wed	Sep 20	More logistic regression			
11	Fri	Sep 22	Multiple Logistic Regression / Multinomial Logistic Regression /Project day			
	Mon	Sep 25	***Review***			
	Wed	Sep 27	**Midterm #1**			
	Fri	Sep 29	No class - Dr Munch out of town			